CEG 4750/6750 (Information Security)
Homework # 4

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Question 1 (15 points). Consider the RSA algorithm. Let the two prime numbers, p=7 and q=5. You need to derive appropriate public key (e,n) and private key (d,n).

(a) Can we pick e=3? If yes, what will be the corresponding (d,n)?
Answer: First of all,

$$n = p \times q = 7 \times 5 = 35$$
$$\varphi(n) = (p-1)(q-1) = 6 \times 4 = 24$$

And for e to be valid, it must be coprime with φ(n): gcd(3, 24) = 3 $\neq$ 1. Therefore, e=3 is not valid.

(b) Can we pick e=5? If yes, what will be the corresponding (d,n)?
Answer: For e=5, we again verify validity by cheking gcd(5,24) = 1, which satisfies the requirement. Meaning, yes, we can pick e = 5. We then compute the private key d such that 5d = 1(mode 24). Now solving this gives d = 5. Therefore, the public key is (e, n) = (5, 35), and the private key is (d, n) = (5, 35).

(c) Use e=5, how to encrypt the number 3? What is the encrypted value?
Answer: Using e = 5, encryption is performed using $c = m^e$ mod n. Substituting m = 3, we compute $c = 3^5$ mod 35 = 243 mod 35 = 33. Hence, the encrypted value of 3 is 33.

Question 2 (10 points, Question for Graduate Students) Is PU=(7, 77) and PR=(43,77) a valid RSA public key and private key pair? Show your work.
Answer: To verify whether PU = (7, 77) and PR = (43, 77) form a valid RSA key pair, we first factor n = 77 = 7 x 11, giving φ(n) = 60. A valid RSA pair must satisfy e.d = 1 (mod φ(n)). Here, 7 x 43 = 301, and 301 mode 60 = 1. Since this condition holds and gcd(7,60) = 1, the given keys are indeed valid.

Question 3 (10 points). Using extended Euclid Algorithm, find the multiplicative inverse of 37 mod 434.
Answer: To find the multiplicative inverse of 37 modulo 434, we apply the extended Euclidean algorithm to solve 37x = 1 (mod 434). Since gcd(37, 434) = 1, an inverse

exists. By working through the algorithm, we obtain x = 317, which satisfies the equation 37 x 317 = 1 (mod 434). Therefore, the multiplicative inverse is 317.

Question 4 (10 points). Show the first eight words of the key expansion for a 128-bit key 0x 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F.
Answer: The given 128-bit key is first divided into four 32-bit words, which form the initial key schedule. AES key expansion then generates additional words by applying transformations such as RotWord (byte rotation), SubWord (S-box substitution), and XOR with round constants. Each new word depends on the previous word and the word four positions earlier, ensuring both diffusion and non-linearity. Thus, the first eight words consist of the original four words directly from the key, followed by four derived words computed through this iterative process.

Question 5 (20 points) Given the plaintext {0x000102030405060708090A0B0C0D0E0F} and the key {0x1111111112121212141414141818181818},

(1) Show the original content of State, displayed as a 4*4 matrix.
Answer: The plaintext is arranged into a 4×4 state matrix in column-major order, meaning bytes are filled column by column rather than row by row. This forms the initial state used in AES transformations. So the original content of state would be

00	04	08	0C
01	05	09	0D
02	06	0A	0E
03	07	0B	0F

(2) Show the value of State after initial AddRoundKey.
Answer: In the AddRoundKey step, the state matrix is XORed with the key matrix. This operation introduces key dependency at the very beginning of the encryption process and ensures that identical plaintext blocks produce different outputs under different keys. Thus, the value of State after initial AddRoundKey is:

11	16	1c	14
10	17	1D	15
13	14	1E	16
12	15	1F	17

(3) Show the value of State after Subbytes.
Answer: During the SubBytes step, each byte in the state matrix is replaced using the AES S-box. This substitution is non-linear and provides strong resistance against linear and differential cryptanalysis by introducing confusion into the data. So the State after Subbytes would be:

82	47	9C	FA
CA	F0	A4	59
7D	FA	72	47

C9	59	C0	F0

(4) Show the value of State after Shiftrows.

Answer: In the ShiftRows step, each row of the state matrix is cyclically shifted by a specific offset: the first row remains unchanged, while subsequent rows are shifted left by increasing positions. This operation helps spread the influence of each byte across multiple columns, contributing to diffusion in the encryption process. Thus, the value of State would be:

82	47	9C	FA
F0	A4	59	CA
72	47	7D	FA
F0	C9	59	C0